

Online Appendix to “Informal Regulation and Housing Supply: Evidence from Chicago’s Aldermanic Privilege”

Jacob Herbstman

A Data and Sample Construction

This appendix provides additional detail on the data sources and sample rules described in Section 2.

A.1 Data Sources

The property data come from Cook County Assessor sales, assessment, parcel, and improvement files. RentHub provides rental listings from 2014–2022. Building permits and ward boundaries come from the City of Chicago. The amenity controls use school locations, Chicago Park District boundaries, major streets, CTA stops, and Lake Michigan geometry; CTA distances use the permanent station network in place during the listing month or on the sale date.

Demographic controls come from American Community Survey block groups and 2010 and 2020 census blocks. I allocate population, households, tenure, race, ethnicity, and education counts to wards according to the share of each block group’s area falling within the relevant ward boundaries. Ward shares use those allocated counts, while ward median household income is the household-weighted mean of block group medians. When historical ward polygons overlap slightly, I scale the intersections to the block group’s area inside the union of Chicago wards so no area is counted twice.

A.2 High-Discretion Permits

The permit records cover Department of Buildings permits since 2006. The high-discretion group contains the permit types defined in Section 2.2; low-discretion placebo permits are the remaining routine types, excluding signs. The event-study outcomes omit records marked cancelled, revoked, or suspended.

Processing time is the number of days between application and issuance. The processing-time regression excludes missing dates, negative durations, and same-day permits; same-day permits remain in the lagged ward-month permit count. I assign each permit to the ward containing its coordinates under the boundaries in force on its application date.

A.3 Permit Event-Study Panel

The treatment panel compares each census block’s ward under the pre-2015 and post-2015 maps. Blocks are projected to EPSG:3435 and assigned by largest-area overlap. Let $a_o(b)$ and $a_d(b)$ denote the aldermen representing census block b under the origin and destination ward maps. Blocks moved to a different ward are classified using the difference between these aldermen’s stringency scores:

$$\Delta S_b = S_{a_d(b)} - S_{a_o(b)}.$$

Positive values indicate assignment toward greater stringency, negative values indicate assignment toward greater leniency, and unchanged blocks form the comparison group.

For this design, stringency scores use permits filed from 2006 through 2014. Both origin and destination scores are assigned using the incumbents serving in 2014. The main sample retains blocks only when both incumbents remained in office when the new map took effect, so treatment matches the change the block experienced without using the outcome of the 2015 election. Online Appendix C.4 adds wards with electoral turnover and interprets the resulting estimate as assigned rather than necessarily realized stringency.

Permits are aggregated by block and application year for event times $t = -5$ through $t = 5$. The sample includes blocks whose polygon centroid is within 500 ft of a pre-redistricting boundary. Treated blocks use the fixed origin-destination boundary created by redistricting; comparison blocks use their nearest incident pre-redistricting boundary. The estimation sample requires at least one high-discretion permit during event times $t = -5$ through $t = -1$.

A.4 New Construction

The construction sample combines Cook County residential improvement records with commercial valuation records for properties assessed under multifamily model groups. I link records across parcel divisions, replacement PINs, and the two Assessor sources so that each project enters once. I then compare the resulting sample with every issued residential new-construction permit, adding completed projects absent from the Assessor files when units, building area, and lot area can be measured consistently.

Projects must have positive lot area, positive building area, at least one dwelling unit, and a construction year from 2006 through 2022. Assessor histories sometimes report different years or repeat whole-building characteristics across assessment cards. I use permit and completion dates to select the year that best matches completion, combine cards that clearly describe one building, and remove later PINs representing the same project. For 52 projects citywide, including 26 within 500 ft of a ward boundary, the administrative records require manual resolution; the replication files include a spreadsheet recording each decision.

Locations first come from the Historical Parcel Universe in the construction year. When the original PIN is unavailable there, I use the current location of that PIN or a verified historical address. Each project retains one location, preventing predecessor and successor PINs from entering separately. Multifamily status primarily follows Assessor property class. When the reported unit count does not plausibly describe the whole building, permit descriptions and other property records determine the whole-building count, with the

remaining decisions recorded in the same spreadsheet.

Each project receives the broad zoning group in force on June 15 of its construction year. I combine official City zoning maps from 2012, 2014, and 2016 with passed map amendments dating to 2006, use annual parcel maps to follow sites whose PINs change, and retain planned developments as a separate zoning group. The reconstruction covers every project in the main and placebo boundary samples and reproduces the classifications observed in later official zoning maps. FAR is total building area divided by lot area; DUPAC is dwelling units divided by lot acreage. Projects are assigned to the ward map and alderman in office on June 15 of the reported construction year because the Assessor does not provide an exact completion date. All ward assignment and distance calculations use EPSG:3435, with distance reported in feet.

A.5 New-Construction Summary Statistics

Table 1 describes the 3,692 projects within 500 ft used in both main density regressions, including 822 multifamily projects.

A.6 Home Sales

The sales sample covers Chicago transactions from 1999–2022, with the analysis restricted to 2006–2022. It includes residential classes 202–211 but excludes mixed-use properties and condominiums. Sales must exceed \$10,000, use a warranty or trustee deed, include recorded and non-identical seller and buyer names, and involve a single parcel; land-only transactions are excluded. Prices are deflated to 2022 dollars using the monthly Chicago CPI-U and winsorized at the 1st and 99th percentiles of the analysis sample.

I geocode each sale by matching its full PIN to the Historical Parcel Universe in the sale year, falling back to the 2025 location of the same PIN when necessary. Ward and alderman assignments use the exact sale date: transactions before May 18, 2015, use the old map, and those on or after that date use the new map. Signed distances are calculated in EPSG:3435

and oriented by relative aldermanic stringency.

A.7 RentHub Listings

The rental data begin as scrape-level listings. I standardize addresses and building types and treat zero, blank, NA, N/A, NULL, NONE, and UNKNOWN addresses as missing. Because the data do not identify units, a property proxy combines normalized address and rounded coordinates, using coordinates alone when address is missing. A floorplan proxy adds bedrooms, bathrooms, square feet, and building type. I collapse repeated scrapes to the median rent for each floorplan and scrape date and then to the median daily rent for each floorplan-month. Before monthly aggregation, rent and square footage are trimmed at the 1st and 99th percentiles within year-by-bedroom cells.

The sample requires positive rent, valid coordinates, and assignment to a Chicago ward. Ward and alderman assignments use the listing's first observed date, with May 18 as the map transition date. The main analysis retains only reliable locations: it excludes manually rejected coordinates, unstable address clusters, addressless listings concentrated at generic coordinates, and cases where the address's modal coordinate changes the ward, neighboring ward, ward pair, or measured boundary distance by more than 100 ft. Regressions also require square footage, bathrooms, and amenity controls. Bedrooms enter categorically, so studios remain in the sample.

B Alderman Fixed Effects and Index Validation

This appendix provides additional detail on how the aldermanic stringency index is constructed and validated.

B.1 Construction of the Stringency Index

Aldermanic stringency is measured using processing times from the high-discretion permit sample described in Section A.2. The processing-time regression uses positive-duration permits through December 2022. Same-day permits remain in the lagged ward-month permit count.

Stage 1: Adjusting Processing Times. I regress permit-level log processing time on characteristics of the permit’s location (distance to the CBD, distance to the lake, and nearby CTA stations), ward demographic controls (median household income, homeownership rate, population, and racial composition), lagged ward-month permit volume, and fixed effects for year-month, permit type, and review type. Appendix A describes how block group demographics are allocated across ward boundaries to construct the ward controls.

Stage 2: Alderman Differences. I average the adjusted processing times by ward and month, then regress those ward-month averages on alderman indicators $\alpha_{a(w,m)}$ and the same lagged permit-volume control. Ward-months are weighted by their permit counts. Here $a(w,m)$ denotes the alderman serving ward w in month m . The alderman fixed effect captures the systematic component of processing time attributable to the alderman.

Empirical Bayes Shrinkage. Raw alderman fixed effects are shrunk toward the grand mean using an empirical Bayes procedure: $\hat{\alpha}_a^{\text{EB}} = B_a \hat{\alpha}_a$, where $B_a = \hat{\omega}^2 / (\hat{\omega}^2 + \hat{\sigma}_a^2)$ is the shrinkage factor and $\hat{\omega}^2$ is the estimated between-alderman variance.

Index Standardization. The final stringency index, S_a , is standardized to have mean zero and unit standard deviation across aldermen. Higher values indicate more stringent aldermen.

Table B.1 reports the first regression used to construct the stringency index. The dependent variable is permit-level log processing time. The specification controls for observed

differences across wards and permit locations, as well as the mix of permit and review types. The purpose of this stage is to remove systematic differences in permit timing attributable to those observed characteristics before estimating the alderman-specific component. Panel B of Figure B.1 shows the distribution of the final stringency scores.

B.2 Validation of the Stringency Index

The stringency index is intended to capture alderman-specific regulatory frictions rather than purely ward-level conditions. Panel C of Figure B.1 compares predecessor and successor aldermen within the same ward. The weak relationship indicates that the index is not simply reflecting time-invariant ward characteristics.

Table B.1: First-Stage Processing-Time Regression

	Log Processing Time (1)
Median HH Income (\$10k)	-0.015 (0.016)
Share Black	0.175 (0.278)
Share Hispanic	0.401 (0.272)
Share White	0.529 (0.352)
Homeownership Rate	0.262 (0.222)
Population (10k)	-0.003 (0.020)
Dist. to CBD (km)	-0.017*** (0.006)
Dist. to Lake (km)	-0.002 (0.005)
CTA Stations (800m)	-0.009*** (0.003)
Lag Permits	-0.002*** (0.0006)
Year $\times$ Month FE	✓
Permit Type FE	✓
Review Type FE	✓
N	120,520

Notes: Permit-level OLS regression used to construct the stringency index, using permit records through December 2022. Covariates are median household income, racial shares, homeownership, population, distance to the CBD, distance to the lake, CTA station density, and lag permit volume. Fixed effects are listed in the table. Regressions are unweighted. Standard errors are clustered by ward and reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

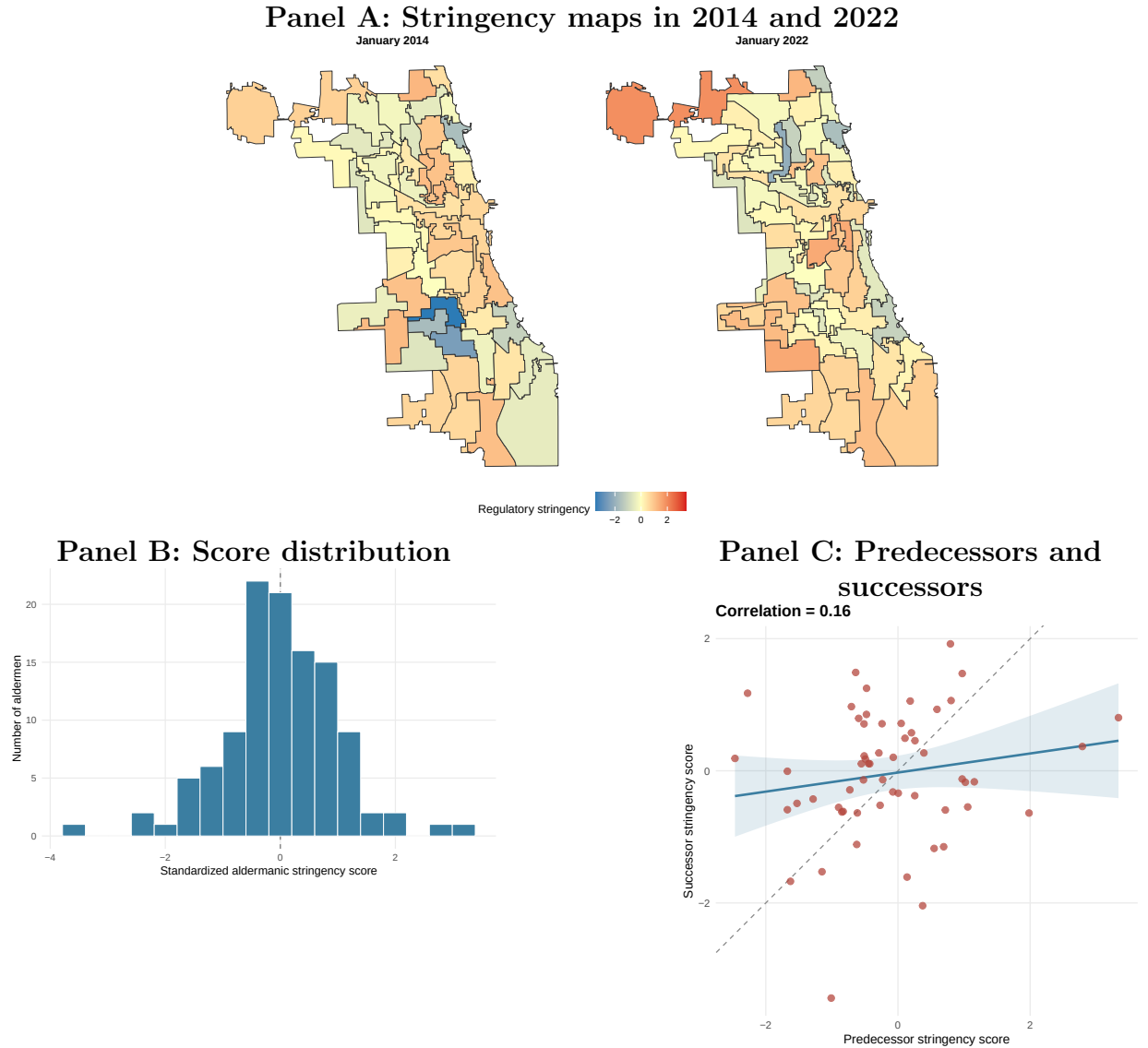


Figure B.1: Geography, Distribution, and Within-Ward Persistence of Aldermanic Stringency

Notes: Stringency scores are constructed from residualized permit processing times through December 2022 with empirical Bayes shrinkage and standardized to mean zero and unit standard deviation. Panel A maps the scores of aldermen serving in January 2014 and January 2022 onto the corresponding ward boundaries; both maps use the same color scale, with warmer shades indicating more stringent aldermen. Panel C compares each alderman's score with that of a predecessor or successor who represented the same ward in an adjacent period.

C Additional Permit Evidence

This appendix reports additional permit evidence that complements the descriptive discussion in Section 2 and the event study in Section 4.1.

C.1 Balance Before Redistricting

Table C.1 compares blocks that changed wards in 2015 with nearby blocks that remained in the same ward. Before the new map took effect, the two groups were broadly similar in permit activity and neighborhood characteristics. The joint test across all listed characteristics has a p -value of 0.333.

Table C.1: Pre-Redistricting Balance Between Reassigned and Unchanged Blocks

Variable	Control	Treated	Difference	SE	p -value	N
<i>Permit history</i>						
High-discretion permits, 2010–2014	2.668	2.649	0.086	0.081	0.291	2,340
Low-discretion permits, 2010–2014	4.278	4.377	0.146	0.412	0.725	2,340
No low-discretion permit, 2010–2014	0.114	0.127	0.008	0.023	0.713	2,340
New-construction permits, 2010–2014	0.381	0.349	0.013	0.044	0.766	2,340
No new-construction permit, 2010–2014	0.773	0.798	-0.008	0.013	0.525	2,340
<i>Neighborhood characteristics</i>						
Median household income	58,210	53,228	-1,352	1,354	0.324	2,324
Homeownership rate	0.404	0.407	0.000	0.011	0.998	2,340
Bachelor’s degree or higher share	0.441	0.399	-0.000	0.009	0.970	2,340
Black population share	0.301	0.272	0.003	0.006	0.667	2,340
Hispanic population share	0.226	0.276	-0.011	0.009	0.230	2,340
Median gross rent	1,068	1,022	-19	16	0.246	2,285
Median home value	309,400	283,787	-8,561	11,019	0.441	2,288
Average household size	2.550	2.694	-0.004	0.052	0.932	2,340
Median age	33.8	33.5	0.1	0.4	0.798	2,340
Population per square kilometer	8,350	8,187	-100	378	0.791	2,340

Notes: Treated blocks were reassigned to a different ward by the 2015 ward map; control blocks remained in their original ward. The sample requires the 2014 origin- and destination-ward incumbents to remain in office when the new map took effect. It also requires at least one high-discretion permit during 2010–2014. The sample contains 687 treated and 1,653 control blocks within 500ft of a ward boundary in 44 ward pairs containing both groups and is constructed before estimation. The event study can also compare blocks reassigned in opposite directions and therefore includes additional blocks. PPML drops blocks and ward-pair-year cells with no permits for the outcome being studied. Permits are grouped by application year. Neighborhood characteristics are 2014 ACS five-year block-group estimates. Difference is the coefficient on the treated indicator from a regression with ward-pair fixed effects; standard errors are clustered by ward pair. Means are unadjusted block-level means. The joint-test p -value for all listed covariates is 0.333 (complete-case $N = 2,249$).

C.2 Permit Event-Study Estimates

Table C.2 reports pooled estimates for years 0 through 5 after redistricting. It shows the main model, which treats moves toward greater stringency and greater leniency as opposite changes, and a more flexible model that estimates each direction separately. Low-discretion permits show a more muted response.

Table C.2: Permit Effects with Combined and Separate Reassignment Directions

	High-Discretion	Low-Discretion
Main estimate	-0.127** (0.049)	-0.017 (0.050)
<i>Directions estimated separately</i>		
Assigned toward more stringent	-0.231** (0.096)	0.037 (0.074)
Assigned toward more lenient	0.067 (0.059)	0.043 (0.065)
One-half difference between directions	-0.149*** (0.052)	-0.003 (0.048)
Equal-and-opposite test p -value	0.180	0.432
Block fixed effects	Yes	Yes
Ward-pair $\times$ year fixed effects	Yes	Yes
Positive pre-period permit activity required	Yes	Yes
N	36,830	35,541

Notes: Poisson pseudo-maximum likelihood estimates use census-block-year observations from 2010–2020 within 500 ft of a ward boundary. Stringency scores are estimated using permits filed from 2006 through 2014. The sample retains blocks whose 2014 origin- and destination-ward incumbents remained in office when the new map took effect. It also requires at least one high-discretion permit during event times $t = -5$ through $t = -1$. Blocks are classified as assigned toward greater stringency, assigned toward greater leniency, or unchanged under the new map. The main estimate constrains assignments toward greater stringency and greater leniency to have equal-and-opposite effects in logs. The second model estimates the two assignment effects separately. For the model that estimates each direction separately, the reported comparison is one-half of the difference between the more-stringent and more-lenient coefficients. The equal-and-opposite test evaluates whether the two separately estimated coefficients sum to zero. Each model includes block fixed effects and ward-pair $\times$ year fixed effects. Fixed-effect PPML drops blocks with no outcome in any sample year and ward-pair-year groups with no outcome. Standard errors, in parentheses, are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Allowing the Two Reassignment Directions to Differ

Panels B and C of Figure 2 allow moves toward greater stringency and greater leniency to have different effects. Table C.2 shows that one-half of the difference between those two effects is nearly identical to the main estimate.

Panel A of Figure C.1 shows the corresponding path for low-discretion permits. The response is more muted than for high-discretion permits.

C.4 Broader Assigned-Change Sample

Panel B includes blocks from wards that elected new aldermen in 2015. Treatment remains based on the 2014 origin- and destination-ward incumbents, so the estimate captures assigned rather than necessarily realized changes in stringency. The annual number of high-discretion permits is 8.3% lower, a somewhat smaller estimate than in the main sample.

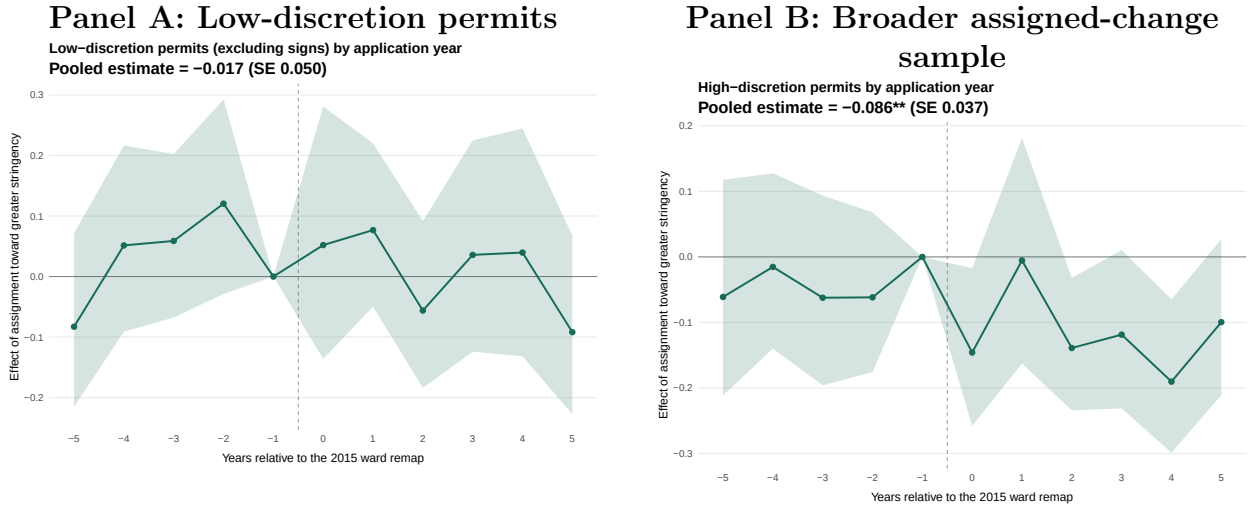


Figure C.1: Additional Permit Event-Study Checks

Notes: Panel A uses low-discretion permits excluding signs; its pooled log coefficient is -0.017 (SE = 0.050), with pre-trend $p = 0.066$. Panel B adds blocks from wards with aldermanic turnover while retaining treatment based on the 2014 incumbents; its pooled coefficient is -0.086 (SE = 0.037), with pre-trend $p = 0.665$. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

C.5 Permit Summary Statistics and Volume

Table C.3 compares the high- and low-discretion permit groups and reports their ward-month processing-time and permit-volume correlations. High-discretion permits are much slower than lower-discretion permits: the mean processing time is 57.3 days versus 26.2 days, and the median is 30 days versus 6 days. They also display more cross-alderman variation in processing times. Ward-months with longer high-discretion processing times have fewer high-discretion permits. Low-discretion permits show a weak positive relationship instead.

Panel A: Processing-time summary statistics

Statistic	High-Discretion	Low-Discretion
Total permits in sample	121,420	124,133
Mean processing time (days)	57.3	26.2
Median processing time (days)	30.0	6.0
Alderman-level mean of processing time	60.3	26.6
Alderman-level IQR of mean processing time	24.5	13.5
Number of unique wards	50	50
Number of unique aldermen	114	114

Panel B: Ward-month processing time and permit volume

Permit group	Corr. of mean processing time and ward-month permit volume
High-Discretion	-0.123
Low-Discretion	0.106
All	-0.021

Table C.3: Summary Statistics for High- and Low-Discretion Permits

Notes: The sample uses positive-duration permit records with applications begun through December 2022. The permit groups are defined in Section 2.2. The alderman-level interquartile range is computed across alderman-specific mean processing times. Panel B reports Pearson correlations across ward-month cells.

D Additional Density RD Checks

This appendix checks whether the density result appears at false cutoffs, reflects fixed differences in location, or depends on unusual boundary geometry or projects closest to the boundary. Unless otherwise noted, each estimate compares the two 100 ft bands immediately across the ward boundary using Equation (4).

D.1 Stringency-Score Checks

The first check asks whether a project's own permits influence the score used to classify its boundary. For each project, I recalculate the scores for the two aldermen after removing permits matched to that project. The estimates are effectively unchanged. The second check drops boundaries where the two alderman scores are close. All four point estimates remain negative when the minimum gap is either 0.25 or 0.50 standard deviations, although the multifamily estimates become less precise as the sample shrinks.

Table D.1: Density Estimates Under Alternative Score Classifications

	All Construction		Multifamily	
	Log(FAR)	Log(DUPAC)	Log(FAR)	Log(DUPAC)
<i>Panel A: Removing permits linked to each project</i>				
Current score	-.129*** (.048)	-.172** (.067)	-.149** (.069)	-.194* (.099)
N	3,692	3,692	822	822
Ward pairs/segments	124/849	124/849	87/415	87/415
Score excluding the project's permits	-.129*** (.048)	-.172** (.067)	-.149** (.069)	-.194* (.099)
N	3,692	3,692	822	822
Ward pairs/segments	124/849	124/849	87/415	87/415
<i>Panel B: Excluding boundaries with similar scores</i>				
Minimum score gap: 0.25 SD	-.097* (.050)	-.140* (.076)	-.086 (.077)	-.097 (.103)
N	2,774	2,774	686	686
Ward pairs/segments	119/751	119/751	78/356	78/356
Minimum score gap: 0.50 SD	-.117** (.056)	-.160* (.086)	-.166 (.101)	-.188 (.130)
N	2,158	2,158	518	518
Ward pairs/segments	100/569	100/569	64/272	64/272

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The second row removes permits matched to each construction project before recalculating the alderman scores used for that project. The minimum-gap rows retain boundaries where the absolute difference between the two scores is at least the stated number of standard deviations. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.2 Placebo Cutoffs

Figure D.1 moves the cutoff 1000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority, and none of the eight nearest-band estimates is statistically distinguishable from zero.

D.3 Continuity in Location Characteristics

Table D.2 tests whether fixed features of a project’s location change at the ward boundary. None of the differences in distance to downtown, the nearest school, the nearest park, or Lake Michigan is statistically distinguishable from zero in either sample.

Table D.2: Continuity in Location Characteristics at Ward Boundaries

	All Construction	Multifamily
Distance to downtown (miles)	-0.001 (0.010)	-0.009 (0.012)
Distance to nearest school (miles)	0.005 (0.005)	-0.004 (0.009)
Distance to nearest park (miles)	-0.005 (0.006)	-0.010 (0.010)
Distance to Lake Michigan (miles)	0.004 (0.007)	-0.003 (0.011)
Observations	3,692	822

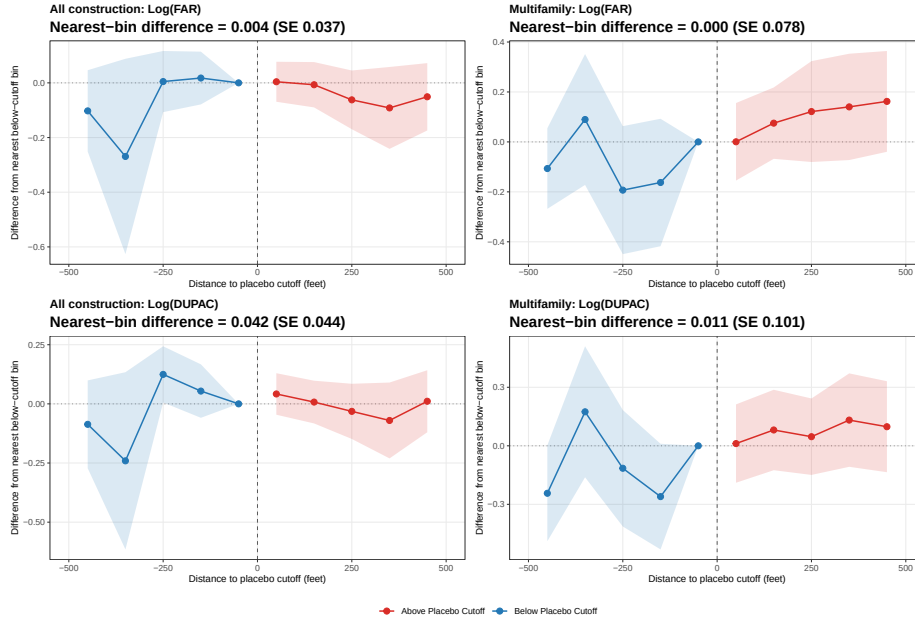
Notes: Each row uses the listed location characteristic as the dependent variable. The reported coefficient compares the 0–100 ft band on the more-stringent side with the –100 to 0 ft band on the less-stringent side. Regressions include 100 ft distance-band indicators, boundary-segment fixed effects, and construction-year fixed effects. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.4 Boundary Geometry

Some ward boundaries follow expressways, waterways, arterial roads, or other physical features. Table D.3 therefore repeats the analysis after restricting the sample based on the geometry of the assigned boundary segment. Removing segments with substantial expressway or water overlap leaves the estimates almost unchanged. A broader physical-feature

Panel A: 1000 ft inside the less-stringent side

Placebo cutoff 1,000 feet inside the less-stringent ward



Panel B: 1000 ft inside the more-stringent side

Placebo cutoff 1,000 feet inside the more-stringent ward

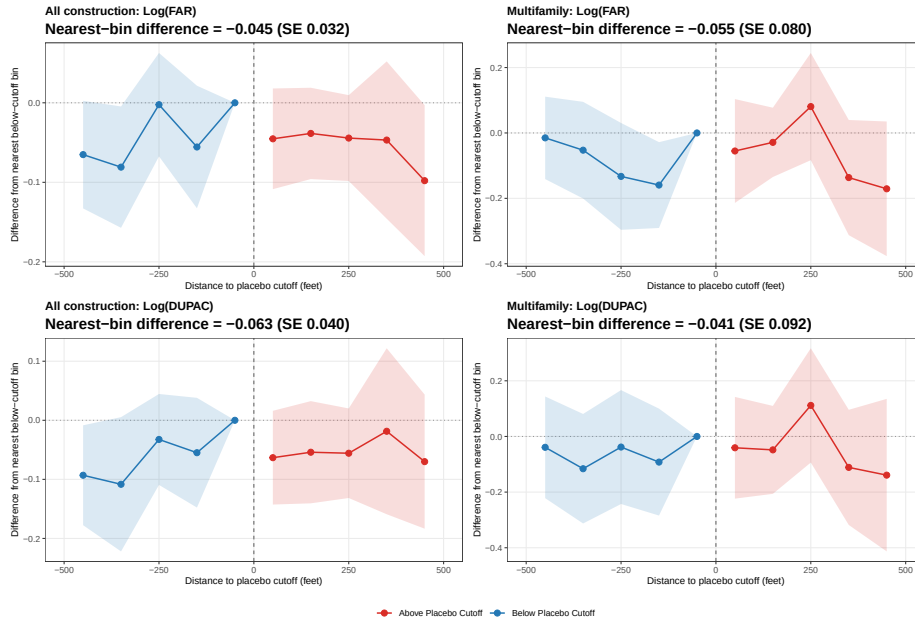


Figure D.1: Density Estimates at Placebo Cutoffs

Notes: Each artificial cutoff lies 1000 ft inside the indicated side of the true ward boundary. The model uses a 500 ft window around the artificial cutoff and the same 100 ft distance bands, controls, and fixed effects as Equation (4). The nearest band below the artificial cutoff is omitted and normalized to zero; subtitles report the difference in the nearest band above it. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

and arterial-road restriction attenuates the multifamily estimates, while the all-construction estimates remain negative. Restricting the sample to locally straight boundaries also produces four negative estimates. These estimates are generally smaller and less precise in the reduced sample, but they do not suggest that boundary curvature alone creates the main pattern.

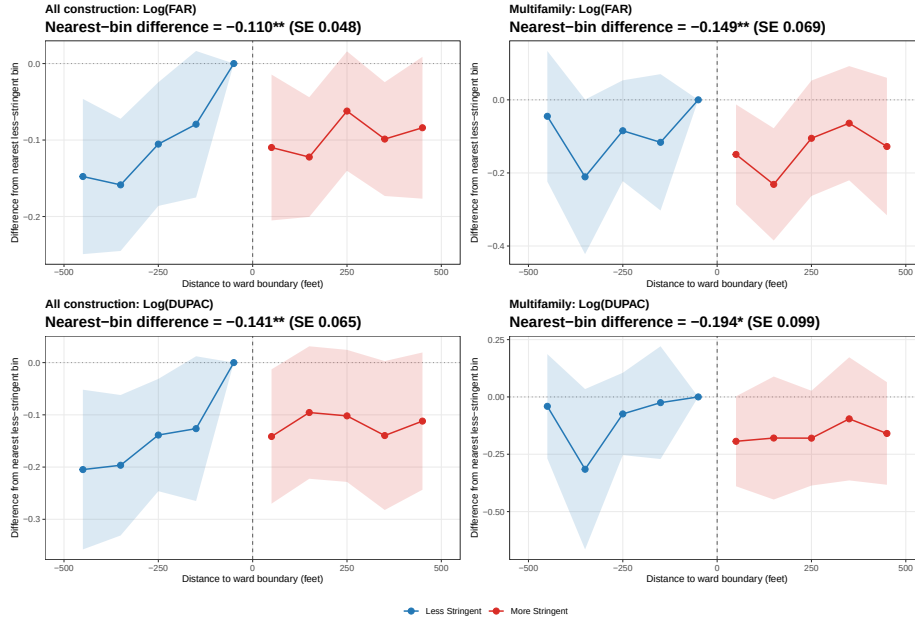
Table D.3: Density Estimates Under Boundary-Sample Restrictions

	All Construction		Multifamily	
	Log(FAR)	Log(DUPAC)	Log(FAR)	Log(DUPAC)
Main sample	-0.129*** (0.048)	-0.172** (0.067)	-0.149** (0.069)	-0.194* (0.099)
Observations	3,692	3,692	822	822
Limited expressway or water overlap	-0.128*** (0.047)	-0.171** (0.067)	-0.149** (0.069)	-0.194* (0.099)
Observations	3,443	3,443	799	799
Limited physical-feature or arterial overlap	-0.118** (0.046)	-0.160** (0.062)	-0.112 (0.070)	-0.106 (0.088)
Observations	2,888	2,888	660	660
Straight boundary segment	-0.085 (0.052)	-0.125** (0.059)	-0.096 (0.082)	-0.173* (0.101)
Observations	2,877	2,877	592	592

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The expressway-or-water restriction removes segments for which either feature overlaps at least half of the segment. The broader restriction removes segments with at least 50% park, water, or cemetery overlap, 40% expressway overlap, or 75% arterial-road overlap. A boundary is locally straight when both endpoints of a 328 ft line tangent to the project's nearest boundary point fall within 49 ft of the actual boundary. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Excluding projects within 25 ft

True ward boundary, excluding projects within 25 feet



Panel B: Excluding projects within 50 ft

True ward boundary, excluding projects within 50 feet

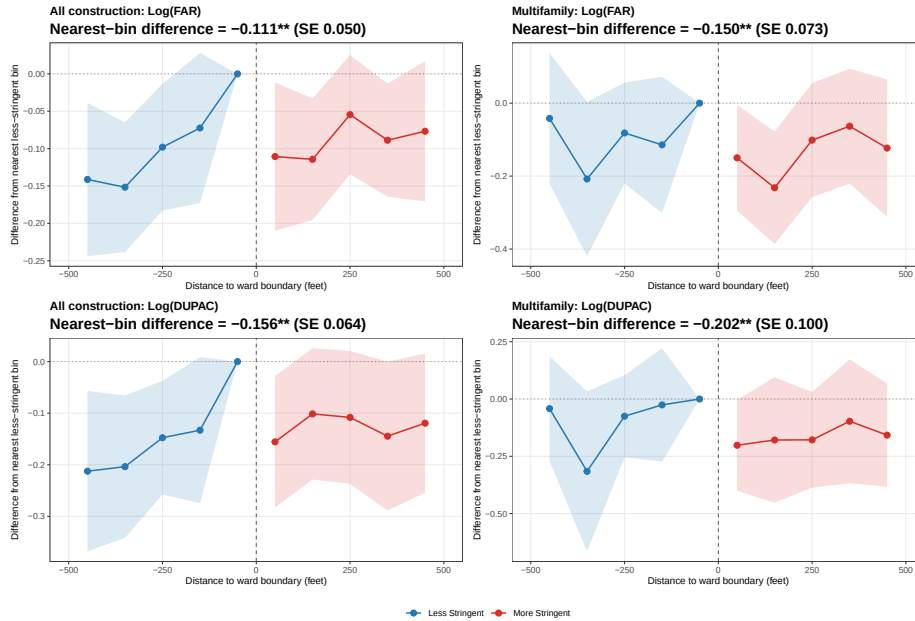


Figure D.2: Density Estimates Excluding Projects Nearest the Boundary

Notes: This check asks whether projects immediately adjacent to a boundary drive the main result. Each panel excludes projects within the indicated distance and otherwise uses the 500 ft sample and Equation (4). After each exclusion, the remaining portion of the -100 to 0 ft band forms the omitted category. In each panel, all four nearest-band estimates remain negative. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

E Additional Price Boundary Checks

Shifted-cutoff price placebos

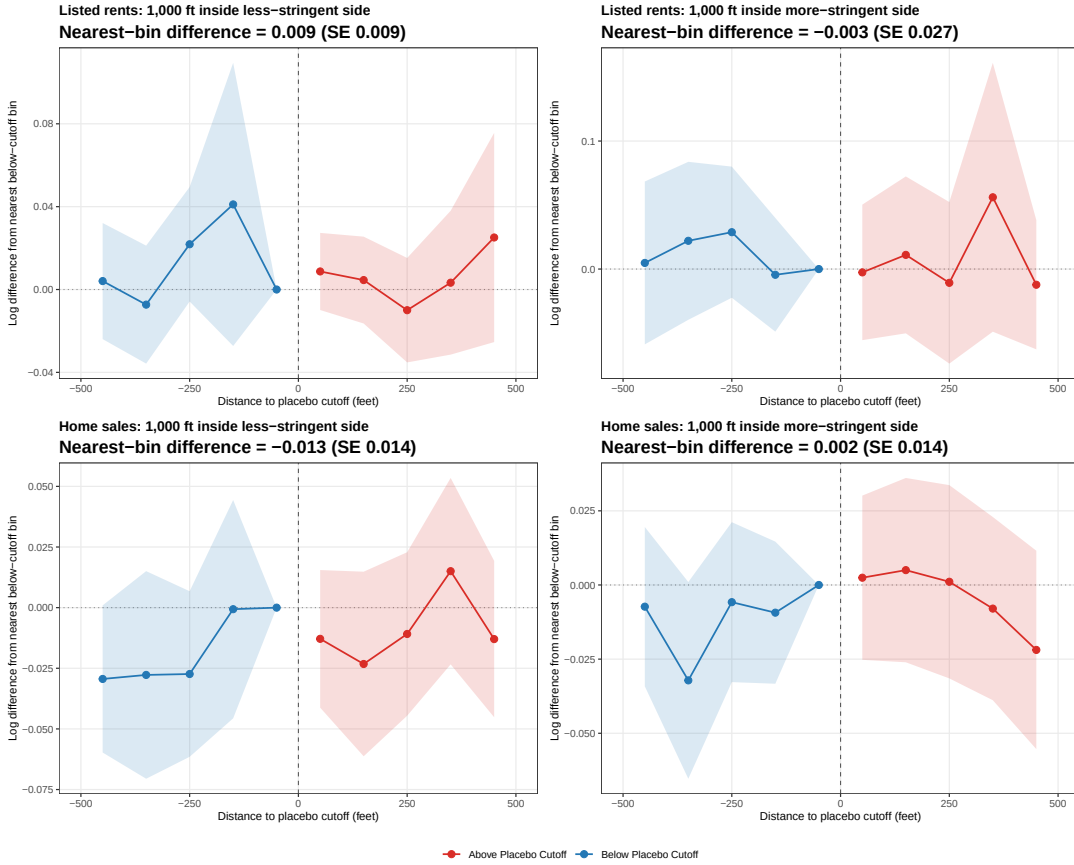
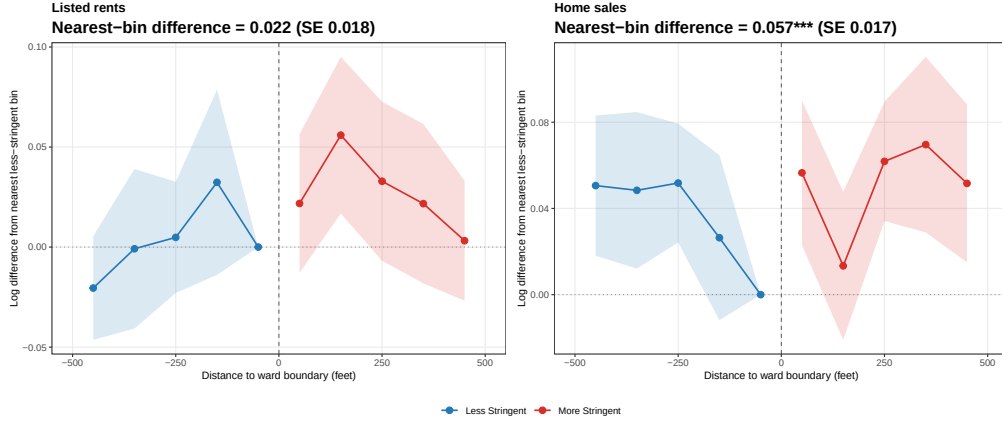


Figure E.1: Placebo Cutoffs for Listed Rents and Home Sale Prices

Notes: The placebo windows do not overlap the main 500 ft sample, and none of the nearest-band estimates is statistically distinguishable from zero. The left column places an artificial cutoff 1000 ft inside the less-stringent ward; the right column places it 1000 ft inside the more-stringent ward. The top row reports listed rents and the bottom row reports home sale prices. Each model uses a 500 ft window around the artificial cutoff and otherwise follows the corresponding specification in Figure 4. Each point is the estimated log difference between a 100 ft distance band and the band immediately below the artificial cutoff. Shaded areas are 95% confidence intervals. Standard errors are clustered by ward pair.

Panel A: Locally straight boundaries

Price estimates near locally straight ward boundaries



Panel B: Donut samples

Price estimates excluding observations nearest the boundary

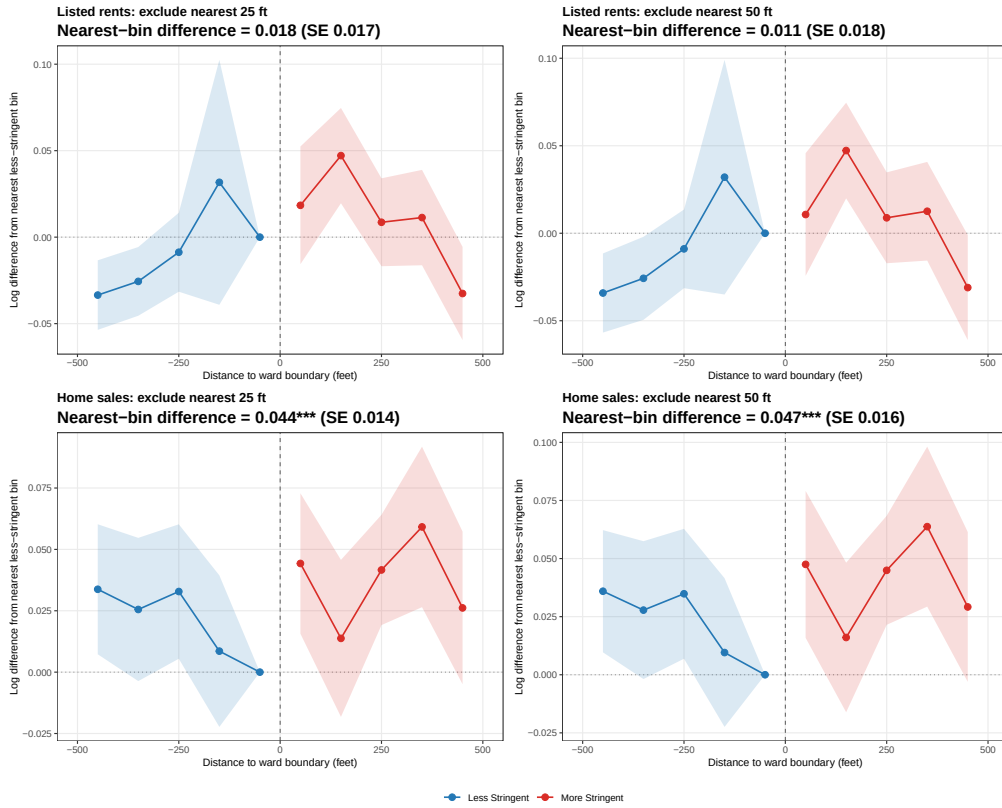


Figure E.2: Additional Price-Boundary Robustness Checks

Notes: Panel A restricts the sample to locally straight boundaries; its estimates are somewhat larger than the full-sample results. It retains locations where both endpoints of a 328 ft line tangent to the nearest boundary point lie within 49 ft of the actual boundary. In Panel B, the left column excludes observations within 25 ft and the right column excludes observations within 50 ft; the top row reports listed rents and the bottom row home sale prices. The donut estimates are similar to the full-sample results. All models otherwise follow Figure 4. Shaded areas are 95% confidence intervals, and standard errors are clustered by ward pair.